UNIT - I

INTRODUCTION TO TQM

Quality - Dimensions - TQM - Quality Planning - Quality Costs - Basic Concepts

1. Definition of Quality

Quality = fitness for use (Juran). The product or service should satisfy the needs of the customer at a price he is willing to pay.

- Latin word "**qualitas**" meaning "nature of a thing" or "degree of excellence".
- Crosby: Quality is **conformance to requirements** (zero defects).
- Deming: Quality is **a predictable degree of uniformity and dependability at low cost,** suited to the market.
- Feigenbaum: Quality is the total composite product and service characteristics of marketing, engineering, manufacturing and maintenance through which the product meets the expectation of the customer.
- ISO 9000:2000: Quality is the **degree to which a set of inherent characteristics fulfils requirements.**
- Taguchi: Quality is the loss imparted to society from the time the product is shipped.
- Customer's view: **quality = meeting or exceeding customer expectations.**

Quality can be viewed in 5 ways

- **Transcendent view:** quality is innate excellence, recognised only by experience (e.g. a Rolls Royce).
- **Product based view:** quality is a measurable quantity - more

features/ingredients means higher quality.

- **User based view:** "fitness for use" - depends on who the user is.
- **Manufacturing based view:** conformance to specifications, prevent defects, reduce rework.
- **Value based view:** performance at an acceptable price, quality + cost together.

Why is quality important?

- Customer satisfaction and loyalty, repeat business.
- Competitive advantage, global market share.
- Lower cost - less scrap, rework and warranty claims.
- Company reputation and image, higher profit.
- Product liability - poor quality can lead to legal action.

> Remember: Quality is not an expense, it is an investment. Quality is everybody's job.

2. Dimensions of Quality

Two sets are commonly written. Garvin's 8 dimensions for **products** and the 5 dimensions (RATER) for **services.**

A. Garvin's eight dimensions of product quality

1. **Performance:** primary operating characteristics of a product. Eg. picture clarity of a TV, speed of a car.
2. **Features:** secondary characteristics that supplement the basic function. Eg. remote control, Bluetooth in a car.
3. **Reliability:** probability that the product will not fail within a given time. Measured by MTBF (mean time between failures).
4. **Conformance:** degree to which design and operating characteristics meet established standards/specifications.
5. **Durability:** amount of use before the product deteriorates or must be replaced - the product life.

6. **Serviceability:** speed, courtesy, competence and ease of repair; after-sales service.
7. **Aesthetics:** how the product looks, feels, sounds, tastes or smells - personal judgement.
8. **Perceived quality:** reputation, brand image, advertising - indirect measures of quality.

B. Dimensions of service quality (SERVQUAL - Parasuraman)

Dimension	Meaning	Example
Tangibles	physical facilities, equipment, appearance of staff	clean bank branch, uniforms
Reliability	perform promised service dependably and accurately	order delivered on time
Responsiveness	willingness to help, prompt service	quick reply to a complaint
Assurance	knowledge, courtesy, ability to inspire trust	doctor's confidence
Empathy	caring, individual attention to customer	personalised attention

Other service dimensions

- Time (waiting time), Timeliness, Completeness, Courtesy, Consistency, Accessibility and convenience, Accuracy, Responsiveness.

Exam tip: write Garvin's 8 as the mnemonic "P F R C D S A P" (Performance, Features, Reliability, Conformance, Durability, Serviceability, Aesthetics, Perceived quality).

3. Definition of Total Quality Management

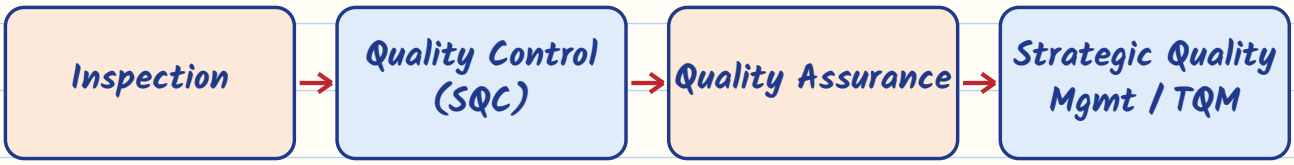
- **Total:** involves everyone in the organisation - top management to workers, suppliers and customers.
- **Quality:** meeting customer requirements, fully and every time.

- **Management:** planning, organising, leading and controlling - quality has to be managed, not left to chance.

TQM is a management approach of an organisation, centred on quality, based on the participation of all its members and aiming at long-term success through customer satisfaction and benefits to all members of the organisation and to society. (ISO 8402:1994)
TQM is both a **philosophy** and a set of guiding principles that represent the foundation of a continuously improving organisation. It integrates fundamental management techniques, existing improvement efforts and technical tools under a disciplined approach focused on continuous improvement. (Besterfield)

- Another simple definition: TQM = **TQC (Total Quality Control) + Quality Circles + Continuous improvement + Employee involvement.**

Evolution of quality (quick timeline)



- **Inspection** (1920s): sorting good from bad after production. Detect defects.
- **Quality control** (1930-50s): statistical methods, sampling, control charts (Shewhart).
- **Quality assurance** (1950-70s): prevent defects by system - documents, audits, quality manuals.
- **TQM** (1980s onwards): whole organisation, customer focus, culture of continuous improvement.

Difference: Traditional management vs TQM

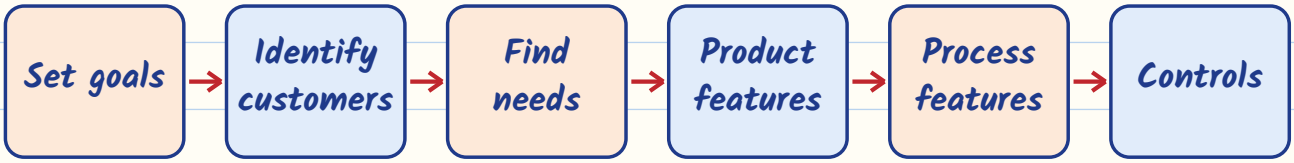
Aspect	Traditional	TQM
Focus	Product, profit	Customer satisfaction
Quality responsibility	QC department	Everyone
Approach	Detect and correct	Prevent defects
Improvement	Occasional	Continuous
Supplier	Adversary, many	Partner, few
Decision making	Intuition	Data and facts
Employees	Cost, followers	Asset, empowered teams

4. Quality Planning

Quality planning is the activity of establishing the quality objectives and requirements, and of defining the operational processes and related resources needed to fulfil them. (Juran: the first process of the Juran Trilogy)

Juran's steps of quality planning (6 steps)

1. **Establish quality goals** - what is to be achieved.
2. **Identify the customers** - internal and external.
3. **Discover customers' needs** - stated, real, perceived, cultural, unintended uses.
4. **Develop product features** that respond to the needs.
5. **Develop process features** capable of producing those features.
6. **Establish process controls** and transfer the plan to operations.



Quality planning for a product (practical stages)

- Market research - customer need, competitor study.
- Design and development - specifications, prototypes, design review.
- Process planning - machines, tools, methods, inspection plan.
- Supplier planning - material specs, vendor approval.
- Quality manual, work instructions, check sheets.
- Training and resource planning.
- Feedback and corrective action planning.

Benefits of quality planning

- Fewer design changes and customer complaints.
- Better use of resources, less waste and rework.
- Clear objectives, so everybody knows the target.
- Shorter time to market, lower life-cycle cost.
- Quality becomes proactive and not reactive.

5. Quality Costs

Cost of Quality (COQ) is the sum of costs incurred in preventing poor quality, appraising quality and the costs due to failure (internal and external). It is the cost of doing things wrong - and of ensuring they are done right.

Four categories of quality cost (PAF model)

Prevention cost → **Appraisal cost** → **Internal failure** → **External failure**

a) Prevention costs (cost of doing it right first time)

- Quality planning, training of employees, quality circles.
- Supplier evaluation, process control, preventive maintenance.
- Design reviews, pilot runs, quality improvement projects.

b) Appraisal costs (cost of checking)

- Incoming material inspection, in-process and final inspection.
- Testing, calibration of instruments, laboratory, quality audits.
- Cost of inspection staff and test equipment.

c) Internal failure costs (defects found BEFORE delivery)

- Scrap, rework/repair, re-inspection, retest.
- Downtime due to defects, yield losses, material review, failure analysis.

d) External failure costs (defects found AFTER delivery)

- Warranty claims, returns and replacements, complaint handling.
- Product recall, field service, product liability, penalties.
- **Loss of customer goodwill and future sales** (the hidden cost).

Prevention + Appraisal = **Cost of Good Quality (conformance).**
Internal + External failure = **Cost of Poor Quality (non-conformance).**

The iceberg

- Visible costs (scrap, rework, warranty) are only about **10 to 20%** of the real cost.
- Hidden costs: lost customers, lost sales, overtime, expediting, extra inventory, low morale, management time. These form the big part below the water.

Optimum quality cost model

- Prevention and appraisal cost **increase** as the quality of conformance (defect-free level) rises.
- Failure costs **decrease** as quality improves.
- Total cost curve is U-shaped - minimum at the optimum point. Modern view: spend more on prevention and the total cost keeps falling - 100% good is economical (zero defect view).

Quality level	Prevention + appraisal	Failure cost	Total

Low	Low	Very high	High
Optimum	Moderate	Moderate	Minimum
Very high (modern view)	Prevention emphasised	Very low	Still falling

- Typical figures: COQ is 15-30% of sales in poor organisations; world-class firms keep it below 5-10%.

6. Analysis of Quality Costs

Quality cost analysis tells where the money is being lost, so improvement effort can be targeted.

Steps of analysis

1. Decide the cost categories and identify the elements in each.
2. Collect data from accounts, inspection and rework records, warranty reports.
3. Prepare the **quality cost report** - category-wise total.
4. Compare with a **base** - sales, manufacturing cost, direct labour, units produced.
5. Study trends month to month and interpret.
6. Select projects with the highest cost (Pareto) and act.

Typical bases for comparison

- **Sales:** COQ / net sales x 100
- **Manufacturing cost:** COQ / total manufacturing cost x 100
- **Direct labour hours or cost,** and **units of product.**

Example - analysis table (Rs. lakh)

Category	Cost	% of total
Prevention	4	8%
Appraisal	10	20%
Internal failure	16	32%
External failure	20	40%

Total COQ	50	100%

- Reading: 72% is failure cost - firm is a "firefighter". Needs to move money to prevention.
- Rule of thumb: ideal ratio about Prevention 50-60%, Appraisal 20-25%, Failure 15-20% of COQ.
- Good indicators: failure cost falling, prevention rising, COQ % of sales falling.

7. Techniques for Quality Costs

- **Pareto analysis:** arrange cost elements in decreasing order; 80% of cost comes from 20% of causes - tackle the vital few.
- **Trend analysis:** plot quality cost against time, sales or production to check improvement.
- **Ratio analysis:** COQ as % of sales, of production cost, of labour, per unit.
- **Cause and effect / Fishbone diagram** to trace the reason for the highest failure cost.
- **Process cost approach:** (ASQ) cost of conformance + cost of non-conformance for each process.
- **ABC (activity based costing) of quality:** charge costs to the activities that cause them.
- **Taguchi loss function:** loss to society even within the specification (Unit IV).
- **Quality cost budgeting and control:** set target, compare actual, variance report.
- **Flow charts:** identify where the costs arise in the process.
- **Quality cost reports:** monthly management summary, departmental reports, project reports.

Uses / Benefits of quality cost analysis

- Gives a common language - money - to talk with top management.
- Identifies improvement opportunities and justifies investments in prevention.
- Measures the effect of the quality programme.
- Helps in budgeting and in resource allocation.

Limitations

- Hidden costs are difficult to measure; data collection is expensive.
- Needs a good accounting system; different definitions across firms.

8. Basic Concepts of TQM

> TQM rests on the idea that quality is built in during design and production, not inspected in at the end. It is a long term, company-wide effort.

Main elements

- **Customer focus:** the customer decides whether quality is good. Know the needs, measure satisfaction.
- **Leadership:** top management creates vision, values, policy and provides resources.
- **Employee involvement and empowerment:** teams, quality circles, suggestion schemes, training.
- **Continuous process improvement:** PDCA, Kaizen - small and never ending steps.
- **Process approach:** results come from controlled processes.
- **Supplier partnership:** suppliers are part of the quality chain.
- **Fact based decisions:** data, SPC and the seven tools.
- **Prevention not detection:** "do it right the first time" (Crosby).
- **Benchmarking and performance measures.**
- **Systems approach:** integrate all the processes towards one aim.

The TQM wheel/pillars



Principles in short

- Internal customer concept - each person receives from one and supplies another. "Next process is the customer".
- Zero defects, Right first time.
- Quality is a never-ending journey, not a destination.
- Cost of quality - spend to prevent, not to fix.
- Open communication, trust and teamwork across departments.

Obstacles (preview of Unit II)

- Lack of top management commitment, resistance to change, poor planning, wrong training, short-term thinking.

Key terms to remember

Term	Meaning
Quality assurance	Planned actions to give confidence that requirements are met
Quality control	Operational techniques to fulfil requirements
Quality circle	Small voluntary group solving work problems
Zero defects	Crosby's performance standard
Internal customer	Next person/department in process

Important Exam Questions (Unit I)

Short answer (2 marks)

- Define quality and TQM.
- List the 8 dimensions of quality.

- What are the SERVQUAL dimensions?
- Define cost of quality. What are the four components?
- What is the Pareto principle?

Long answer (10 marks)

1. Explain the dimensions of quality with suitable examples.
2. Define TQM. Explain the basic concepts and principles of TQM.
3. Explain quality planning. Discuss Juran's quality planning steps.
4. Describe the elements of quality cost. How are quality costs analysed? Explain the optimum cost model.
5. Compare the traditional management approach with TQM.

Revision line: Quality = customer satisfaction. TQM = everyone + everything + continuously. COQ = Prevention + Appraisal + Failure.

UNIT - II

HISTORICAL REVIEW

Quality Council - Statements - Strategic Planning - Deming - Barriers - Benefits - Leader - Gurus - Cases

1. Historical Review of Quality (background)

- **Pre-1900:** craftsmen made and inspected their own products; guilds set standards.
- **1900s - Taylor:** scientific management; separation of planning from doing; inspectors appointed.
- **1924 - Walter Shewhart** (Bell Labs): control chart, statistical quality control.
- **1940s - World War II:** military standards, sampling inspection (MIL-STD-105).
- **1950 - Deming and Juran** go to Japan (JUSE); Japanese industry adopts SQC and company-wide quality.
- **1960s - Zero Defects** (Crosby, Martin Co.); quality circles (Ishikawa, Japan 1962).
- **1970s - Japanese quality** (Toyota, TPS, Kaizen) captures world markets.
- **1980s - TQM** emerges in USA and Europe; 1987 Malcolm Baldrige National Quality Award; ISO 9000 series issued.
- **1990s onward** - Six Sigma (Motorola 1987, GE), QS 9000, ISO 14000, EFQM model.
- **2000s** - ISO 9001:2000 process approach; today ISO 9001:2015, Industry 4.0 and digital quality.

Core idea: quality moved from INSPECTION -> CONTROL ->

ASSURANCE -> TQM.

2. Quality Council

The **Quality Council** (also called Quality Steering Committee) is the group of senior managers who provide overall direction, guide and monitor the TQM effort of the organisation.

Composition

- Chairman: **Chief Executive Officer (CEO)** or top-most executive.
- Members: heads of production, marketing, finance, HR, design, purchase etc. and a representative of the union where relevant.
- The **Quality Officer / Quality Manager** acts as the co-ordinator (secretary).
- In large firms there are sub-councils at division and plant level.

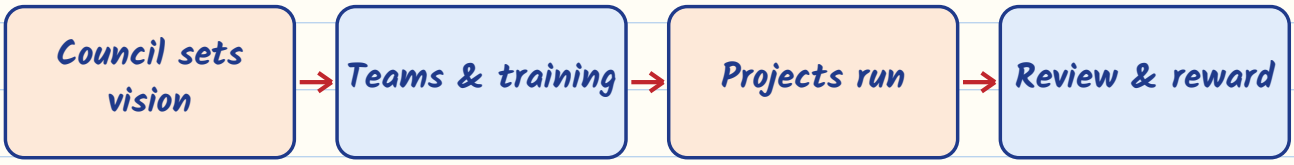
Duties and responsibilities

- Develop and review the **vision, mission and quality policy.**
- Estimate the cost of poor quality; set quality goals and strategy.
- Establish the TQM implementation plan - schedule, resources, budget.
- Set up and support **quality teams** (cross functional teams, quality circles).
- Review progress against goals regularly (monthly or quarterly).
- Approve training programmes; recognise and reward quality achievement.
- Remove barriers, communicate to all, and monitor customer satisfaction.
- Develop new measures when the old ones become inadequate.

Why a council is needed

- Shows visible top management commitment, gives continuity and coordination.

- Ensures that quality efforts in different departments do not conflict.



3. Quality Statements

Quality statements communicate the purpose and direction of the organisation. The three are **Vision, Mission and Quality Policy,** along with values/guiding principles.

a) Vision statement

> A short declaration of what the organisation wants to **become** in the future - its dream and long-term direction.

- Concise, inspiring, easy to remember, future oriented, customer centred.
- Eg. "To be the world's most trusted supplier of automotive components." / Eg. Microsoft: "A computer on every desk and in every home."

b) Mission statement

> The **purpose of existence** of the organisation - who the customers are, what is provided and how.

- Answers: Who are we? Who are our customers? What do we do? How do we do it better than others?
- Should be clear, brief (one or two sentences), specific, understood by all employees.
- Eg. "To provide reliable and affordable power products to meet or exceed our customers' expectations through continuous improvement."

c) Quality policy statement

> The overall intention and direction of an organisation related to quality as formally expressed by top management (ISO 9000).

- Must be **appropriate to the purpose** of the organisation; commit to meeting requirements and continual improvement of QMS.
- Provides a framework for setting quality objectives; communicated and understood at all levels; reviewed for suitability.
- Eg. Ford: "Quality is Job 1."
- Guidelines for writing: equal emphasis on customer, employee and supplier; short; the employees should be able to say it from memory; reflects top management's commitment.

d) Core values / guiding principles

- Customer first, integrity, respect for people, teamwork, continuous learning, social responsibility.

Statement	Question answered	Time frame
Vision	Where do we want to go?	Long term future
Mission	Why do we exist?	Present and ongoing
Quality policy	How do we commit to quality?	Operational, reviewed regularly

4. Strategic Planning

> **Strategic quality planning** is the process of establishing long-range (3 to 5 years) objectives and the strategies to achieve them, with quality built into the business plan - so that the customer needs are met and the organisation grows.

- Strategic planning is led by the **top management** and the quality council; it connects the vision to the day to day actions.

Seven steps of the strategic quality planning process (Besterfield)

1. **Customer needs:** find the needs and expectations through surveys, focus groups, complaints.
2. **Customer positioning:** where does the firm stand relative to competitors - benchmarking, SWOT.
3. **Predict the future:** technology, market, legal, economic trends.
4. **Gap analysis:** gap between the present position and the vision.
5. **Closing the gap:** choose strategies, develop the action plan.
6. **Alignment:** align the strategies with the mission, vision and values of the organisation.
7. **Implementation:** deploy objectives to departments, teams and individuals; allot resources.



Strategic planning terms

- **SWOT** - Strengths, Weaknesses, Opportunities, Threats.
- **Goals** - broad, long term; **Objectives** - specific, measurable, time bound (SMART).
- **Policy deployment (Hoshin Kanri):** the top level strategic goals are cascaded to every level; each manager draws his own objectives and means; reviewed periodically (catchball process).
- **Balanced scorecard** (Kaplan and Norton): financial, customer, internal process, learning and growth perspectives.

Strategic planning model - key inputs

- Mission, vision, customer data, competitor data, employee feedback, supplier input, past performance, regulatory requirements.

- Outputs: long term objectives, short term action plans, resource plan, performance measures.

5. Deming Philosophy

- **W. Edwards Deming** (1900-1993), American statistician, "father of the Japanese quality revolution". Deming Prize in Japan (1951).
- Philosophy: **quality is the responsibility of management;** 85% of problems arise from the system (management), only 15% from workers.
- Deming stressed reduction of **variation;** 'Chain reaction' - improve quality -> costs decrease (less rework) -> productivity up -> capture market -> stay in business -> provide more jobs.

Deming's 14 points for management

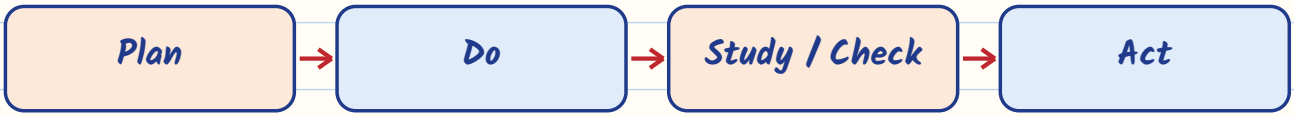
1. **Create constancy of purpose** for improvement of product and service.
2. **Adopt the new philosophy** - we are in a new economic age; do not accept defects.
3. **Cease dependence on inspection** to achieve quality; build quality in.
4. **End the practice of awarding business on price tag alone;** move to single supplier, long term relationship.
5. **Improve constantly and forever** the system of production and service.
6. **Institute training on the job.**
7. **Institute leadership** - help people do a better job; supervision, not policing.
8. **Drive out fear** so everyone works effectively for the company.
9. **Break down barriers between departments** - teamwork.
10. **Eliminate slogans, exhortations and targets** for the workforce (zero defects posters).

11. **Eliminate numerical quotas** and management by objectives; use leadership.
12. **Remove barriers that rob** workers of pride of workmanship; abolish annual rating/merit system.
13. **Institute a vigorous programme of education** and self-improvement.
14. **Put everybody to work** to accomplish the transformation - top management action.

Deming's seven deadly diseases

- Lack of constancy of purpose; emphasis on short term profits; evaluation by performance, merit rating or annual review; mobility of management (job hopping); running a company on visible figures only; excessive medical costs; excessive costs of warranty (fuelled by lawyers who work on contingency fee).

Deming cycle (PDCA/PDSA)



- Plan the change, Do it on a small scale, Study the results, Act - adopt, adapt or abandon. Repeat.

Deming's "System of Profound Knowledge"

- Appreciation for a system; knowledge of variation; theory of knowledge; psychology (human behaviour).

6. Barriers to TQM Implementation

- **Lack of management commitment** - quality treated as a slogan; no time and money given.
- **Inability to change organisational culture** - resistance to change, old habits, fear of job loss.
- **Improper planning** - no clear aims, no timeline, no sponsors.

- **Lack of continuous training and education;** wrong training at wrong time.
- **Incompatible organisational structure** and isolated departments - departmental silos.
- **Ineffective measurement techniques** and lack of data and results.
- **Inadequate use of empowerment and teamwork;** middle managers feel threatened.
- **Inadequate attention to internal and external customers.**
- **Poor communication** and lack of trust between management and employees.
- **Failure to recognise and reward** - no motivation.
- Short term thinking, expecting quick results, "flavour of the month" attitude.
- Paying inadequate attention to suppliers; copying other firms' programmes without adapting.

Overcoming the barriers

- Visible leadership, communication of vision, training at all levels, pilot projects with quick wins, involve unions, measure and publish progress, reward teams, review by quality council.

7. Benefits of TQM

Tangible benefits

- Improved product quality and reliability; fewer defects, rework and scrap.
- Lower cost and higher productivity; reduced cost of quality.
- Reduced customer complaints, warranty and returns.
- Increased sales, market share and profit; better delivery performance.
- Shorter cycle times and lower inventory.

Intangible benefits

- Higher customer satisfaction and loyalty; better company image.
- Improved employee morale, motivation and job satisfaction; teamwork and communication.
- Better supplier relationships; culture of continuous improvement.
- Better decision making with data; improved safety and environment.
- Competitive advantage; easier ISO certification, entry into global markets.

> Benefits flow: Quality up -> cost down -> productivity up -> customer satisfied -> profit up.

8. Characteristics of a Successful Quality Leader

- Gives **personal attention** to customers and quality; visible and active, "walks the talk".
- Has a clear **vision** and communicates it; sets the direction and creates the values.
- Builds **trust and respect;** honest, fair, ethical.
- Believes in **people:** delegates, empowers, trains, coaches and listens.
- Encourages **teamwork** and removes interdepartmental barriers; avoids blame - drives out fear.
- Fact based decisions; uses data and statistical thinking.
- Customer driven: knows customers, supplier partnership.
- Continuous learner, an agent of change, takes risks, accepts mistakes as learning.
- Recognises and rewards achievement.
- Long term thinking, patient, persistent; builds a quality culture; focus on prevention.

Leadership styles in TQM

Style	Feature
Autocratic	Leader decides alone - not suited to TQM
Participative	Involves team in decisions - preferred
Servant/transformational	Inspires change, supports people

9. Contributions of Gurus of TQM

1. W. Edwards Deming

- 14 points, PDCA cycle, 7 deadly diseases, variation (common vs special causes), Profound knowledge, statistical methods.

2. Joseph M. Juran

- "Fitness for use". **Juran Trilogy:** quality planning, quality control, quality improvement. Pareto principle (vital few, trivial many). Cost of quality. Book: Quality Control Handbook. 10 steps to quality improvement.

3. Philip B. Crosby

- "Quality is free". **Four absolutes:** quality = conformance to requirements; system = prevention; performance standard = zero defects; measure = price of non-conformance (PONC).
- **14 steps** to quality improvement. Quality vaccine. Quality maturity grid (uncertainty, awakening, enlightenment, wisdom, certainty).

4. Armand V. Feigenbaum

- **Total Quality Control** (TQC), "hidden plant", quality is everybody's work, the cost of quality, "quality at the source", 19 steps.

5. Kaoru Ishikawa

- **Cause and effect diagram** (fishbone), quality circles, 7 QC tools, company wide quality control (CWQC), internal customer "the next process is your customer".

6. Genichi Taguchi

- **Loss function,** robust design, design of experiments (orthogonal arrays), signal to noise ratio, off-line quality control.

7. Walter A. Shewhart

- Control charts, statistical process control, common and assignable cause, original PDCA.

8. Shigeo Shingo

- **Poka-yoke** (mistake proofing), SMED (single minute exchange of dies), zero quality control, just in time.

9. Taiichi Ohno and others

- Toyota production system, kanban, 7 wastes (muda). Masaaki Imai - **Kaizen.**

Guru	Main contribution
Deming	14 points, PDCA
Juran	Trilogy, fitness for use
Crosby	Zero defects, 14 steps
Feigenbaum	TQC
Ishikawa	Fishbone, QC circles
Taguchi	Loss function
Shingo	Poka-yoke

Juran vs Deming vs Crosby (comparison)

Point	Deming	Juran	Crosby
Definition	Predictable uniformity	Fitness for use	Conformance
Approach	Statistics, variation	Planning, control, improve	Zero defects
Responsibility	85% management	Management	Management
Standard	Continuous improvement	Cost of quality	Zero defects

10. Case Studies

Case 1: Toyota Production System

- Problem: low resources after WWII, need for high quality with low cost.
- Action: Just in time, jidoka (autonomation), kaizen, andon cord, quality circles, supplier partnership.
- Result: world leader in quality, very low defects, flexible production.
- Lesson: culture of continuous improvement and respect for people.

Case 2: Motorola (Six Sigma, 1987)

- Problem: field failures and high warranty cost; Japanese competition.
- Action: Six Sigma target of 3.4 defects per million, top management involvement, training of "black belts".
- Result: savings of billions of dollars, Baldrige award 1988.

Case 3: Xerox (Benchmarking)

- Problem: losing market share to Japanese copier makers in the 1980s.
- Action: benchmarking against Fuji Xerox, Canon and L.L.Bean; leadership through quality; employee teams.
- Result: improved quality and market share, Baldrige award 1989.

Case 4: Ford - "Quality is Job 1"

- Deming's advice adopted; employee involvement and supplier partnership; Taurus car success.

Case 5: Tata Steel / Indian examples

- Tata Steel won the Deming Prize (2008); TQM with Kaizen, quality circles. Infosys and Titan follow TQM and Six Sigma.

> How to write a case study answer: Background -> Problem -> TQM tools applied -> Results -> Lessons learned.

Important Exam Questions (Unit II)

Short answer

- What is a quality council? What are its duties?
- Differentiate vision and mission statements.
- List any five of Deming's 14 points.
- Name four barriers to TQM.
- Mention the four absolutes of quality by Crosby.

Long answer

1. Explain the strategic quality planning process.
2. Explain Deming's 14 points and the 7 deadly diseases.
3. Discuss the barriers and benefits of TQM implementation.
4. Describe the characteristics of a successful quality leader.
5. Discuss the contributions of Deming, Juran and Crosby.
6. Explain a case study on TQM implementation (Toyota / Motorola / Xerox).

UNIT - III

TQM PRINCIPLES

Customer Satisfaction - Employee Involvement - Continuous Improvement - Supplier Partnership - Performance Measures

1. Customer Satisfaction

> The customer is the **king** - the one who receives the product or service. Customer satisfaction = the feeling that the product/service has met or exceeded the expectation.

- **Satisfaction = Perception of performance - Expectation.** If P > E the customer is delighted; if P < E dissatisfied.
- Types: **External customer** (buys/uses the product) and **Internal customer** (next person/department in the process).
- Customer is the one who defines quality - "voice of the customer".
- A satisfied customer tells 3-5 people; a dissatisfied customer tells 10-20 people. It costs 5 times more to get a new customer than to keep an old one.

Customer perception of quality

- Quality is **what the customer perceives,** not what the producer thinks.
- Factors that shape perception: performance, features, service, warranty, brand name and reputation, price, advertisement, past experience, word of mouth.
- Perceived quality is affected by: the product, **the service** that goes with it (delivery, installation, help desk), and the **organisation's image.**
- Needs are of 3 kinds: **Basic** (must be there), **Performance** (more is

better) and **Excitement/delighters** (unexpected features) - the **Kano model.**



Kano model (note)

- Basic needs: absence causes dissatisfaction, presence does not satisfy (brakes in a car).
- Performance needs: satisfaction rises with level (mileage).
- Delighters: unexpected - big satisfaction (e.g., free pick-up service).

Ways to understand customers (feedback tools)

- Surveys and questionnaires, focus groups, comment cards, toll free numbers, customer visits, report cards, the internet/social media, mystery shoppers, sales and service staff, warranty data, complaint analysis.
- Customer Satisfaction Index (CSI) - tracking over time; comparing with competitors.

2. Customer Complaints

A complaint is an expression of dissatisfaction. Complaints are **free consulting** - opportunities to improve and to retain the customer.

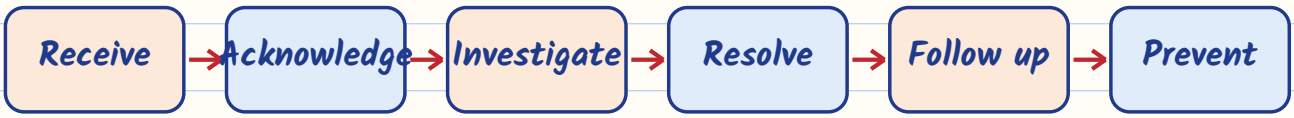
- Only about 4% of unhappy customers complain; the rest leave silently ("iceberg").

Steps to handle complaints effectively

1. **Listen** actively; let the customer finish; do not argue.
2. **Apologise and show empathy;** take responsibility.
3. **Record** the details - what, when, who, where.
4. **Solve** the problem quickly - replace, repair, refund; empower front

line staff.

5. **Follow up** to confirm satisfaction.
6. **Analyse** causes, take corrective action and feed back to design/production.



Best practices

- Make it easy to complain (toll free number, email, app).
- Respond within a fixed time (acknowledge in 24 hours).
- Maintain a complaint database; use Pareto analysis on causes.
- Train employees; reward complaint handling.
- "Service recovery paradox": a well handled complaint can create a more loyal customer than before.

Complaints and the quality system

- ISO 9001 requires monitoring of customer perception and records of complaints and corrective action.

3. Service Quality

- Services are **intangible, inseparable** (produced and consumed together), **heterogeneous** (vary), **perishable** (cannot be stored).
- Hence quality must be **built into the process and the people,** because there is no chance to inspect before delivery.

SERVQUAL gaps model (Parasuraman, Zeithaml, Berry)

- **Gap 1:** customer expectation vs management perception.
- **Gap 2:** management perception vs service quality specification.
- **Gap 3:** specification vs service delivery.
- **Gap 4:** service delivery vs external communication (promises).
- **Gap 5:** expected service vs perceived service (the customer gap) =

result of gaps 1 to 4.

Dimensions (RATER)

- **Reliability, Assurance, Tangibles, Empathy, Responsiveness.**

Translating service quality into practice

- Service blueprinting, standards for response time, moment of truth (every contact), training front-line staff, service guarantee, use of technology (ATM, e-banking), measure through surveys, mystery shopping.
- **Moments of truth** (Jan Carlzon, SAS): each customer-employee contact shapes perception.

Service quality vs product quality

Product	Service
Tangible	Intangible
Can be inspected before delivery	Quality seen during delivery
Stored	Cannot be stored
Produced first, consumed later	Produced and consumed simultaneously

4. Customer Retention

> Customer retention = keeping existing customers satisfied so that they continue to buy. It is the **result of customer satisfaction.**

- Retaining is cheaper than acquiring; loyal customers buy more, pay more, refer others.
- Retention rate = (Customers at end - New customers) / Customers at start x 100.
- **Customer loyalty** is stronger than satisfaction - "completely satisfied" customers are far more loyal than "satisfied" ones.

Factors for retention

- Product quality and consistency, on-time delivery, after-sales service, fair price, quick complaint redress, trust, communication.

Retention strategies

- Loyalty / reward programmes, personalised service, CRM software, customer clubs, warranty and AMC, relationship marketing, customer feedback loop and follow up, delight them beyond expectation.
- Customer lifetime value (CLV) thinking.

Remember: Satisfaction -> Loyalty -> Retention -> Profit.

5. Employee Involvement

Employee involvement means that **all employees participate** in decision-making and improvement; they feel ownership of the quality of their work.

- Employees closest to the job know most about the problems.
- Benefits: better ideas, commitment, reduced resistance to change, higher morale, lower turnover.
- Means: suggestion scheme, quality circles, teams, empowerment, job enrichment, training, recognition.

Motivation

- **Motivation** is the internal drive to act towards a goal. In TQM, managers create conditions in which employees want to do quality work.

Maslow's hierarchy of needs



- Lower needs satisfied -> higher needs drive behaviour. Job security,

belonging in teams, recognition and growth are important in TQM.

Herzberg's two factor theory

Hygiene factors (dissatisfiers)	Motivators (satisfiers)
Salary, working conditions, company policy, supervision, job security	Achievement, recognition, responsibility, the work itself, advancement, growth

- Absence of hygiene factors makes people unhappy; but only motivators can produce satisfaction.

Other theories

- **McGregor's Theory X and Y:** X - people dislike work, need control; Y - people seek responsibility, self-direction. TQM follows Theory Y.
- **Vroom's expectancy theory, McClelland's needs** (achievement, power, affiliation), **Alderfer's ERG.**

How to motivate for quality

- Clear goals, feedback, involvement, fair pay, recognition and rewards, training, job rotation and enrichment, trust and open communication, drive out fear.

Recognition and reward

- Recognition = acknowledgement of effort (thank-you, certificate, notice board); Reward = tangible (bonus, gift, promotion).
- Should be timely, specific, fair, linked to team performance; do not replace with annual merit rating (Deming).

6. Empowerment and Teams

Empowerment is investing people with authority to make decisions and take action in their own area of work, with responsibility and accountability.

Conditions for empowerment

- Everyone understands the need, knows the system and the goal; has training and skills; has information, tools and resources; boundaries are clear; management support, trust; rewards linked.
- Steps: create trust, share information, give training, delegate authority gradually, give feedback.
- Benefits: faster decisions, ownership, motivation, quality at source.

Teams

> A team is a group of people with complementary skills committed to a common purpose, performance goals and approach for which they hold themselves mutually accountable.

- Types of teams: **Process improvement teams, cross-functional teams, natural work teams, self-directed (self-managed) teams, quality circles** (voluntary, 6-12 members, meet regularly), task force / ad-hoc teams, **suggestion** based.

Stages of team development (Tuckman)



Characteristics of successful teams

- Clear mission and goals; open communication; defined roles; trained leader and members; use of problem solving tools; mutual respect and trust; management support; recognition.

Quality circles

- Started in Japan 1962 (Ishikawa/JUSE). Voluntary small group of the same work area; meets 1 hour weekly; **facilitator,** leader, members; identifies, analyses and solves problems; presents to management; seven QC tools.

Problem solving in teams

- Brainstorming, nominal group technique, 5 Why, fishbone, Pareto, PDCA.

Team meeting ground rules

- Agenda, time keeping, minutes, one person at a time, decisions by consensus.

7. Continuous Process Improvement

> Continuous improvement **(Kaizen)** means improving every process, every day, in small steps - with no end.

- Process: a set of activities that converts input into output. Every process has an owner, suppliers and customers; variations can be reduced and wastes removed.
- Basis: **input -> process -> output,** with feedback.

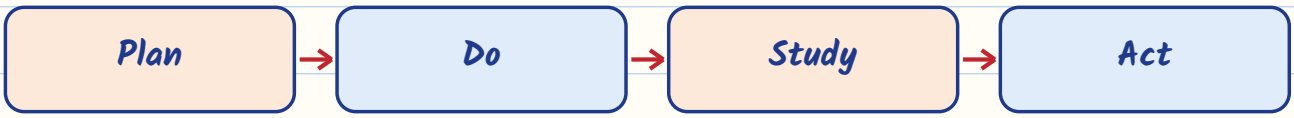
Juran Trilogy



- **Quality planning:** identify customers and needs, design product and process; hand over to operations.
- **Quality control:** operate process, compare actual with goal, act on the difference (keep chronic level and prevent sporadic spikes).
- **Quality improvement:** break through to a new, lower level of chronic waste - project by project.
- Juran diagram: cost of poor quality vs time - planning sets level; sporadic spike is fixed by control; improvement lowers chronic waste.
- Juran's 10 steps to improvement: awareness, goals, organise, train, projects, report progress, recognise, communicate results, keep score,

maintain momentum.

PDSA Cycle (Shewhart/Deming)



- **Plan:** identify the opportunity and plan the change; analyse the present process, collect data, set targets.
- **Do:** implement the change on small scale (pilot).
- **Study (Check):** analyse the results; what did we learn?
- **Act:** if successful standardise and institutionalise; else repeat with new knowledge.
- Repeat continuously - the improvement spiral. A better name than PDCA because "study" stresses learning.
- **Process improvement 7-step:** define the process -> Select target -> Analyse data/causes -> Develop solution -> Implement -> Verify results -> Standardise.

Kaizen

Kaizen = "Kai" (change) + "Zen" (for the better). Masaaki Imai - continuing improvement involving everyone: top managers, managers and workers.

- Kaizen is **people oriented,** small steps, low cost, bottom up as well as top down, process based, and long term. Versus **innovation:** radical, big investment, technology led.
- Kaizen tools: **5S** (Seiri-sort, Seiton-set in order, Seiso-shine, Seiketsu-standardise, Shitsuke-sustain), suggestion system, **Gemba** (go to the actual place), **muda** elimination (7 wastes: overproduction, waiting, transport, over-processing, inventory, motion, defects), standard work, PDCA/SDCA.

- Kaizen events (blitz) - 3-5 day team projects.

Kaizen	Innovation
Small, continuous	Large, one-time
Everyone	Few specialists
Low investment	High investment
Process focus	Result focus

8. Supplier Partnership

> Quality of final product depends on bought-out materials (often 50-80% of cost). Suppliers must be treated as **partners** rather than adversaries.

Partnering

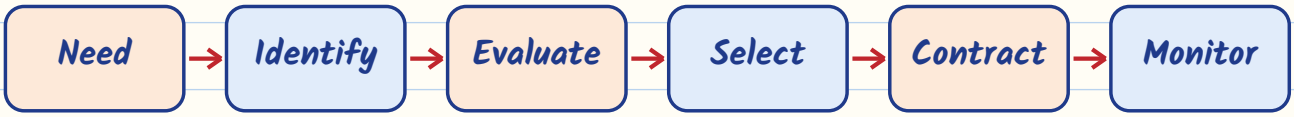
- Long term relationship based on mutual trust, shared goals, shared information and risk; win-win.
- Deming: end the practice of awarding business on price alone; use **single-source** where possible.
- Benefits: lower cost, better quality, JIT delivery, less inspection, joint development, faster innovation.
- Requirements: top management support, mutual training, clear specs, communication, share of improvement gains.

Sourcing

- **Sole sourcing** (one supplier) - good relation, lower cost, but risk; **Multiple sourcing** - risk shared, competition, but less commitment; **Single/dual sourcing** is a compromise.
- Make or buy decision; local vs global; certified suppliers (ISO 9001) require less inspection.

Supplier selection (Besterfield - 9 steps)

1. Determine the need, specification and quantity.
2. Identify potential suppliers (catalogues, trade journals, past records).
3. Short list; send questionnaire/request for quotation.
4. **Evaluate:** financial status, quality system, capacity, technology, location, delivery, service.
5. Visit and audit; sample evaluation.
6. Negotiate the contract; select supplier.
7. Place order; monitor; feedback.
8. Review performance regularly.
9. Develop the supplier.



Supplier rating

- Rating = a numerical score of supplier performance used for ranking, rewarding and decision on future orders.
- Criteria: **Quality** (percentage accepted, rejection ppm), **Delivery** (on time %), **Price/cost, Service/response,** quality system certification.
- Eg. Weighted rating: Quality 40 + Delivery 30 + Price 20 + Service 10 = 100 points.
- Grades: A (preferred) 90+, B (approved) 75-89, C (conditional) 60-74, D (reject) below 60.
- Used for: certification, **vendor of the year** award, reducing inspection of A-class suppliers, corrective action for weak ones.

Relationship development

- Stages: transactional -> cooperative -> partnership -> strategic alliance.

- Actions: supplier conferences, joint training, early supplier involvement in design, long-term contracts, share forecasts, supplier councils, recognition, joint improvement projects, resident engineers.
- Supplier certification (e.g., Ford Q1, ISO 9001, QS 9000).

9. Performance Measures

Basic concepts

> Performance measure = a quantified way of knowing how well the processes and the organisation are performing against goals. "You cannot control what you cannot measure."

- Purposes: identify problems, track progress, drive improvement, motivate, help in decisions, benchmarking.
- Good measures are: **simple, relevant, understandable, timely, accurate, linked to goals, few in number, owned by someone.**
- Measure the right things: customer, process, employee, supplier, financial - not only financial (lagging) but leading indicators.
- Types: **Output** (results) and **process** (leading) measures; Management measures: customer satisfaction, defects ppm, cycle time, on time delivery, cost of quality, absenteeism, training hours, supplier rating.

Strategy for performance measures

- Link to the organisation's strategy; develop measures at each level; use **Balanced Scorecard** - financial, customer, internal business process, learning and growth.
- Steps: identify the process -> define the measure -> set target/standard -> collect data -> analyse -> compare -> take action -> review measure.
- **Quality rating index, Customer satisfaction index, Process capability, Six sigma level, First pass yield, ppm, Return on**

quality.

- Display: control charts, trend charts, dashboards, "scoreboards" displayed where workers can see.

Perspective	Sample measure
Financial	ROI, cost of quality
Customer	CSI, complaints, retention
Internal process	Defects ppm, cycle time, yield
Learning and growth	Training hours, suggestions per employee

Measures for service organisations

- Waiting time, errors per transaction, call abandonment rate, number of complaints, accuracy.

Performance measure case studies

- **Case: Motorola** - used defects per million; cycle time; led to Six Sigma.
- **Case: A bank branch** - measured waiting time, errors, and customer satisfaction; found that long queue at 12-2 pm; added counter and reduced wait from 12 to 5 minutes; satisfaction index up 15%.
- **Case: Xerox** - customer satisfaction, benchmarking market share; improved supplier rating and reduced the number of suppliers from 5000 to 500.
- **Case: Wipro/Infosys** - use balanced scorecard and Six sigma metrics.

Important Exam Questions (Unit III)

Short answer

- Define customer perception of quality.
- Write short notes on the Kano model / the 5 service gaps.
- What is empowerment? What is Kaizen?
- Mention the steps in the PDSA cycle.

- What is supplier rating?

Long answer

1. Explain the customer satisfaction model and the methods of handling complaints.
2. Discuss Maslow and Herzberg's theories and their use in motivating employees.
3. Explain the Juran trilogy with a neat sketch.
4. Explain the PDSA cycle and Kaizen concept with examples.
5. Describe the steps in supplier selection and rating, and the relationship development process.
6. Discuss performance measures - basic concepts and strategy.

UNIT - IV

TQM TOOLS

Benchmarking - QFD - Taguchi - TPM - FMEA - 7 QC Tools - Process Capability - Six Sigma - New 7 Tools

1. Benchmarking

Benchmarking is a systematic and continuous process of measuring and comparing an organisation's products, services and processes against those of the **best-in-class** organisations, to get information that helps to improve performance. (Camp: "the search for industry best practices that lead to superior performance")

- Started at **Xerox** (1979) against Japanese copier makers.
- Benchmark = a reference point / standard of excellence. Benchmark is not copying - it is **learning and adapting.**

Reasons to benchmark

- Stay competitive; set realistic and credible goals based on facts.
- Meet customer requirements, find out what customers expect.
- Break the "not invented here" syndrome - learn from others.
- Find the gaps in the performance and the best practices; quick improvement and fewer mistakes.
- Helps to be proactive, to change the culture, to understand own process better.
- Provides a basis for **continuous improvement** and strategic planning.

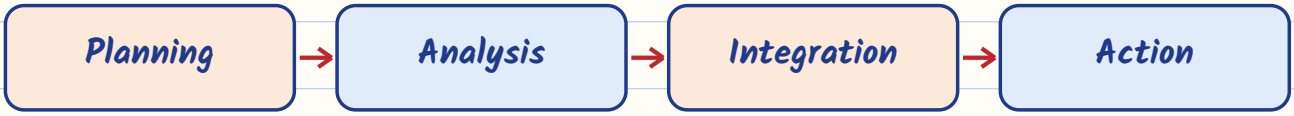
Types of benchmarking

- **Internal** (between units of the same company); **Competitive** (direct competitor); **Functional/Industry** (similar function in different industry); **Generic** (best process irrespective of industry - e.g. Xerox

vs L.L.Bean warehousing).

- Also: performance, process and strategic benchmarking.

Benchmarking process (Xerox 10 steps, grouped into 4 phases)



1. Decide **what** to benchmark (critical success factor).
2. Identify the best competitors / best practices.
3. Identify the **data collection** method and collect the data (surveys, visits, journals, internet).
4. Determine the **current performance gap.**
5. Project future performance levels.
6. **Communicate** the findings and gain acceptance.
7. Establish functional **goals.**
8. Develop **action plans.**
9. **Implement** the plan and monitor the results.
10. Recalibrate benchmarks - continue the cycle.

- Simplified (Besterfield): Plan - Collect - Analyse - Adapt/Improve.

Pitfalls

- Wrong choice of subject; copying blindly; poor partner selection; no top management support; ignoring ethics (a code of conduct: legality, exchange, confidentiality).

Benefits

- Improves quality and productivity; competitive edge; cost reduction; improves customer satisfaction; creates awareness of the best practice.

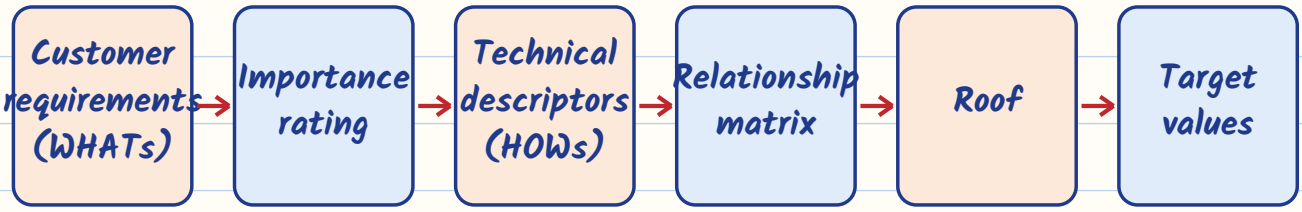
2. Quality Function Deployment (QFD)

QFD is a planning tool that translates the **voice of the customer (VOC)** into technical requirements for each stage of product development and production. Developed in Japan (Yoji Akao, 1966; Mitsubishi Kobe shipyard 1972; used by Toyota).

- Also called the "customer driven engineering" - answers **WHAT** the customer wants and **HOW** the company will provide it.
- Team approach: marketing, design, production, quality, purchase.

House of Quality (HOQ)

- The main matrix of QFD, shaped like a house.



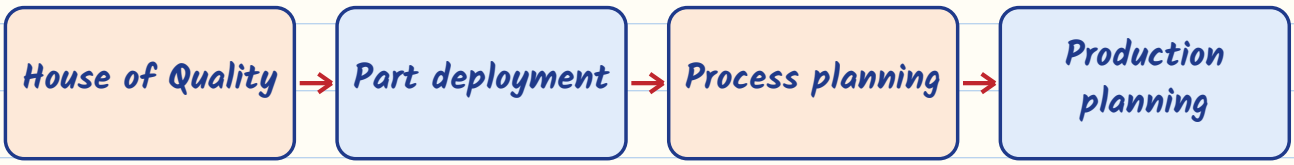
Parts of the House of Quality

1. **Left wall:** customer requirements (WHATs) from market research, with importance ratings.
2. **Ceiling:** technical requirements / design characteristics (HOWs) - measurable engineering responses.
3. **Main body (relationship matrix):** strength of relation between each WHAT and HOW: strong (9, double circle), medium (3), weak (1).
4. **Roof (correlation matrix):** relation among the HOWs - positive (+) or negative (-); to find trade-offs.
5. **Right wall (planning matrix):** customer perception, competitor comparison, sales point, improvement ratio.
6. **Basement (bottom):** technical priorities, competitor technical benchmarking and **target values.**

- Example: Customer wants "easy to open door of car" -> HOW: door

closing effort, seal compression, hinge force.

QFD process (the 4 phases)



- Phase 1: customer needs -> design requirements. Phase 2: design requirements -> part characteristics. Phase 3: part characteristics -> process operations. Phase 4: process -> production/quality control plans.
- Steps: list customer requirements -> rate importance -> list technical descriptors -> relationship matrix -> competitor evaluation -> correlation -> set targets -> select items to deploy.

Benefits of QFD

- Customer focused; reduces design changes and development time (up to 30-50%); lower cost; fewer complaints.
- Improves communication and teamwork; documents the decisions; prioritises resources; competitive benchmarking.
- Quality built in at design stage - **preventive approach.**

3. Taguchi Quality Loss Function

- **Genichi Taguchi** (Japan): Quality loss is the loss imparted to society after the product is shipped. Even if the product is **within specification limits,** any deviation from the **target (nominal) value** causes loss to the customer.
- Traditional view: loss is zero within the tolerance (LSL to USL) and then suddenly jumps (goal-post). Taguchi: **loss increases as a parabola** as the quantity departs from target.

Quadratic loss function

L(y) = k (y - T)^2

- y = actual value of the quality characteristic, T = target value, k = loss coefficient = **A / d^2** (A = cost of failure/scrap at the tolerance limit, d = tolerance half-width).
- For a population: **Average loss = k (sigma^2 + (mean - T)^2)** - loss depends on both variation and off-centring.

Types

- **Nominal-the-best:** L = k (y - T)^2 (e.g., dimension of a shaft).
- **Smaller-the-better:** L = k y^2 (e.g., wear, impurity, noise).
- **Larger-the-better:** L = k (1/y^2) (e.g., strength, mileage).

Example

- A shaft target is 20 mm, tolerance +-0.1 mm. Scrap cost at limits = Rs 50. Then k = 50 / (0.1)^2 = Rs 5000 per mm^2. For a shaft measuring 20.05 mm: L = 5000 x (0.05)^2 = **Rs 12.5.**

Importance / lessons

- Aim for the **target,** not merely inside the limits; reduce variability continuously.
- Basis for robust design, design of experiments (orthogonal arrays), signal-to-noise ratio (S/N).
- Shows quality loss as a system-wide cost - ties to the cost of quality.

Key idea: the closer to the target, the lower the loss - Quality = Uniformity around target.

4. Total Productive Maintenance (TPM)

TPM is a maintenance programme involving **all employees** that aims at maximising equipment effectiveness through a total system of preventive maintenance over the life of the equipment. (Seiichi

Nakajima, Japan, 1971; Nippondenso)

Concept

- Operator = "I operate, I maintain" **(autonomous maintenance);** maintenance staff do planned/skilled work.
- Goals: **zero breakdowns, zero defects, zero accidents.**
- TPM = TQM for equipment. Total = total participation, total effectiveness, total life cycle.

Overall Equipment Effectiveness (OEE)

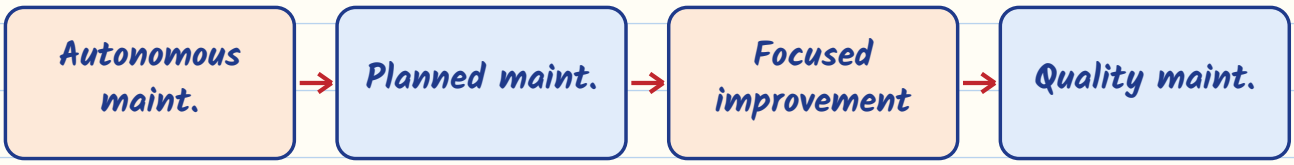
OEE = Availability x Performance rate x Quality rate

- Availability = operating time / planned time; Performance = actual speed / ideal speed; Quality = good pieces / total pieces. World class OEE = 85%.

Six big losses

1. Breakdown, 2. Setup and adjustment, 3. Idling and minor stoppages, 4. Reduced speed, 5. Process defects, 6. Start-up (reduced yield).

Eight pillars of TPM



- Also: Training and education, Early equipment management, Office TPM, Safety Health and Environment.
- Foundation is **5S.**

Improvement needs / steps in TPM implementation

- Management decision and announcement -> education -> create the organisation -> basic policy and goals -> master plan -> kick-off -> build the pillars -> **full implementation and PM award.**
- Needs: top management commitment, training, 5S, planned schedule,

spare-parts management, record of equipment history, small group activities, measure OEE.

Benefits

- Increased productivity and OEE; fewer breakdowns; lower cost; improved safety; better quality; motivated operators.

5. Failure Mode and Effects Analysis (FMEA)

FMEA is a systematic, proactive tool to identify **potential failure modes** of a product/process, their causes and effects, and to prioritise the actions to remove or reduce the risk **before** the failure occurs.

- Types: **Design FMEA (DFMEA), Process FMEA (PFMEA);** also system FMEA, service FMEA. Used in automotive (QS 9000 / IATF), aerospace, medical devices.

Stages of FMEA

1. **Form the team** and define scope; collect drawings, process flow chart.
2. Identify items/functions and list the **potential failure modes.**
3. Describe the **effects** of each failure (on customer, system).
4. Identify the **causes** of each failure mode.
5. List the **current controls** (prevention and detection).
6. Rate **Severity (S), Occurrence (O)** and **Detection (D)** on a 1-10 scale.
7. Calculate **RPN = S x O x D** (Risk Priority Number, max 1000).
8. Recommend **corrective actions** for high RPN; assign responsibility and date.
9. Implement actions, **re-evaluate** the RPN; update the FMEA (a living document).

Item	Failure mode	Effect	Cause	S	O	D	RPN

Brake pad	Wears early	Poor braking	Soft material	9	4	3	108

Rating

- Severity: 1 = no effect, 10 = hazardous without warning. Occurrence: 1 = remote, 10 = very high. Detection: 1 = almost certain to detect, 10 = cannot detect.

Benefits

- Improves reliability and safety; reduces warranty cost and late design changes; documents the knowledge; team based and prevention oriented.

Limitations

- Time consuming; subjective ratings; same RPN can come from different S, O, D combinations.

6. The Seven Tools of Quality (Old / basic QC tools - Ishikawa)

1. **Check sheet** - simple form for collecting data in real time at the place of work (tally marks). Types: defect location, defect cause, frequency check sheet.
2. **Histogram** - bar graph of frequency distribution; shows centre, spread and shape of the data (normal, skewed, bimodal).
3. **Pareto diagram** - bars in descending order with cumulative line; **80:20 rule** - vital few vs trivial many; select the problem to attack first.
4. **Cause and effect diagram (fishbone/Ishikawa)** - brainstorm causes under 6 Ms: Man, Machine, Material, Method, Measurement, Mother nature (environment); effect = head of fish. Use 5-Why for the root cause.
5. **Scatter diagram** - plot two variables to see the relationship

(positive, negative, none); correlation coefficient r.

6. **Control chart** - run chart with the centre line and UCL, LCL (+-3 sigma); separate **common cause** from **special cause** variation. Types: X-bar and R (variable), p, np, c, u (attribute).
7. **Stratification / Flow chart / Run chart** - (depending on the book: stratification, process flow chart) - separating data into groups to find the pattern.



When to use what

Tool	Answers the question
Check sheet	What is happening and how often?
Pareto	Which problem is the biggest?
Fishbone	Why is it happening?
Histogram	How is the data spread?
Scatter	Are two factors related?
Control chart	Is the process stable?

7. Process Capability

Process capability is the ability of a process to produce output **within the specification limits.** Natural process spread = **6 sigma.** Process must be in statistical control first.

- **Cp = (USL - LSL) / 6 sigma** - capability ignoring the centring (potential).
- **Cpk = min [(USL - mean) / 3 sigma , (mean - LSL) / 3 sigma]** - actual capability considering the mean shift.

- Interpretation: Cp < 1 - process not capable; Cp = 1 - just capable (0.27% outside, 2700 ppm); Cp = 1.33 - **capable** (usual minimum); Cp >= 1.67 - very capable; Cp = 2 -> Six Sigma.
- Cpk <= Cp; if Cpk = Cp the process is centred.

Example

- Specification 50 +- 3 mm: USL = 53, LSL = 47. Sigma = 0.8 and mean = 50.4. Cp = 6 / (6 x 0.8) = **1.25.** Cpk = min (2.6 / 2.4 , 3.4 / 2.4) = min (1.08, 1.42) = **1.08** - needs improvement (re-centre and reduce variation).

Steps in a capability study

- Select the characteristic -> collect 25+ subgroups -> check process is in control -> calculate mean and sigma (R-bar/d2) -> compute Cp, Cpk -> take action.

Ways to improve

- Centre the process; reduce variation (machine, material, method); widen tolerance (if permissible); 100% inspection as a last resort.

8. Concept of Six Sigma

Six Sigma is a data driven, customer focused methodology to reduce variation and defects to **3.4 defects per million opportunities (DPMO).** Introduced by Bill Smith at **Motorola (1986-87);** popularised by GE (Jack Welch).

- Sigma level: 1 sigma = 691,462 DPMO; 2 = 308,537; 3 = 66,807; 4 = 6,210; 5 = 233; **6 = 3.4** (with the 1.5 sigma shift allowed).
- DPMO = defects / (units x opportunities per unit) x 1,000,000.

DMAIC (for existing processes)

Define → **Measure** → **Analyse** → **Improve** → **Control**

- **Define** project, customer CTQ; **Measure** current performance; **Analyse** root causes; **Improve** by solution/pilot; **Control** to sustain (control plan, SPC).
- **DMADV / DFSS** for new design: Define, Measure, Analyse, Design, Verify.

Roles ("belts")

- Champion (executive sponsor), Master Black Belt, **Black Belt** (full time project leader), Green Belt (part time), Yellow Belt (team member).

Features

- Top management driven, financial benefit by project, statistical tools, training, focus on CTQ.
- Benefits: lower defects, cost saving, improved satisfaction, culture of discipline.
- Lean Six Sigma = Lean (waste removal) + Six Sigma (variation reduction).

9. New Seven Management Tools

(Used for planning and idea organising - mainly non-numerical data; developed by JUSE 1979.)

1. **Affinity diagram (KJ method)** - group large number of ideas/language data into natural categories.
2. **Interrelationship digraph** - shows the cause and effect relationship among the ideas; arrows show driver and outcome.
3. **Tree diagram** - breaks down a broad goal into progressively detailed levels of tasks (goal -> means).
4. **Matrix diagram** - shows relation between two or more groups of items (L, T, Y, X, C shapes); symbols for strength.
5. **Prioritisation matrices** - rank options by weighted criteria.

6. **Process Decision Program Chart (PDPC)** - map possible problems and countermeasures in a plan.
7. **Activity network diagram (arrow diagram / PERT-CPM)** - sequence and timing of activities, critical path.



Old tools	New tools
Numerical data	Verbal / idea data
Problem solving, control	Planning, strategy
Shop floor	Management

10. Case Studies

Case 1 - Xerox benchmarking

- Compared warehousing with L.L.Bean (generic benchmark): reduced picking time and cost. Result: significant savings and quality leadership.

Case 2 - QFD at Toyota Auto Body

- Rust problems on the car body found from VOC; QFD cut start-up and warranty cost by about 60% over 4 years.

Case 3 - TPM at a manufacturing company

- Problem: frequent breakdown of CNC machines, OEE 55%. Action: autonomous maintenance, 5S, planned PM, operator training. Result: OEE 78%, breakdowns down by 70%.

Case 4 - FMEA in automotive

- Ford (Pinto lesson): failure analysis prior to launch would have found fuel tank risk; now PFMEA is mandatory in supplier systems.

Case 5 - Six Sigma at GE / Motorola

- Savings of billions of dollars; Motorola moved from 3 sigma to near 6 sigma levels.

Important Exam Questions (Unit IV)

Short answer

- Define benchmarking. State the reasons for benchmarking.
- What is the House of Quality? Name its parts.
- Write the Taguchi loss function equation.
- What is OEE? What is RPN?
- Define Cp and Cpk. State the 7 new management tools.

Long answer

1. Explain the benchmarking process with a suitable example.
2. Explain QFD and the house of quality with a neat sketch. State the benefits.
3. Explain Taguchi's loss function and solve a numerical problem.
4. Explain TPM - concept, pillars, six big losses and OEE.
5. Explain the stages of FMEA with an example table.
6. Explain the seven QC tools with sketches.
7. Explain process capability with a numerical example. Discuss Six Sigma and DMAIC.

UNIT - V

QUALITY SYSTEMS

ISO 9000 - Implementation - Documentation - Auditing - QS 9000 - ISO 14000 - Case Studies

1. Need for ISO 9000 and Other Quality Systems

- A **quality system** (Quality Management System - QMS) = the organisational structure, responsibilities, procedures, processes and resources for implementing quality management.
- **ISO** = International Organisation for Standardisation, Geneva (1947). "ISO" comes from the Greek word *isos* = equal. **ISO 9000 series first issued in 1987** (based on British Standard BS 5750 and military standards); revised 1994, 2000, 2008, **2015.**

Why do we need ISO 9000?

- **Customer confidence:** assurance that the supplier has a documented and audited system; no need for customer to audit each supplier (second party audit) - one certificate accepted everywhere.
- **Global market access:** international trade, tender requirement, preferred supplier.
- Consistent quality of product and service; fewer defects and customer complaints.
- Better internal management: clear responsibilities, documented procedures, training.
- Reduces cost of quality, waste and rework; improves productivity.
- Legal and statutory compliance, product liability protection.
- Competitive advantage and improved image and employee morale.
- Starting point and base for TQM (system + continual improvement).

The ISO 9000 family

Standard	Purpose
ISO 9000	Fundamentals and vocabulary
ISO 9001	**Requirements** of QMS (the standard used for certification)
ISO 9004	Guidelines for performance improvement / sustained success
ISO 19011	Guidelines for auditing management systems

- Earlier (1994 version): ISO 9001 (design, production, installation, servicing), 9002 (no design), 9003 (final inspection). In 2000 all merged into **ISO 9001:2000.**

Other quality systems

- **QS 9000** (automotive - Chrysler, Ford, GM), **ISO/TS 16949 -> IATF 16949,** AS 9100 (aerospace), TL 9000 (telecom), ISO 13485 (medical devices), ISO 22000 (food safety), **ISO 14001** (environment), OHSAS 18001 -> **ISO 45001** (health and safety), ISO/IEC 27001 (information security), ISO 17025 (labs), **ISO 26000** (social responsibility).

> Remember: certification is given by an independent **certification body** (registrar); the standard is not a product standard - it is a system standard.

2. ISO 9000:2000 Quality System - Elements

Eight quality management principles (ISO 9000:2000)

1. **Customer focus**
2. **Leadership**
3. **Involvement of people**
4. **Process approach**
5. **System approach to management**
6. **Continual improvement**
7. **Factual approach to decision making**

8. Mutually beneficial supplier relationships



Process model of ISO 9001:2000

- Customers' requirements (input) -> **Management responsibility** -> **Resource management** -> **Product realisation** -> **Measurement, analysis and improvement** -> product -> customer satisfaction (output).
- With the PDCA cycle applied to all processes; feedback of customer satisfaction.

Clauses of ISO 9001:2000 (sections 4 to 8)

Clause	Title	Main requirements
4	Quality Management System	General requirements, documentation: quality manual, control of documents and records
5	Management responsibility	Management commitment, customer focus, quality policy, planning and objectives, responsibility and authority, management representative, management review
6	Resource management	Provision of resources, human resources (competence, training), infrastructure, work environment
7	Product realisation	Planning, customer related processes, design and development, purchasing, production and service provision, control of monitoring and measuring devices

8	Measurement, analysis and improvement	Customer satisfaction, internal audit, monitoring of process and product, control of nonconforming product, data analysis, continual improvement, corrective and preventive action

- (Clauses 1 to 3: scope, normative reference, terms and definitions.)

Key elements explained

- **Quality policy and objectives:** measurable, consistent and communicated.
- **Management review:** top management reviews the QMS at planned intervals - inputs: audit results, customer feedback, process performance, status of corrective actions; outputs: improvement decisions and resource needs.
- **Design control:** planning, inputs, outputs, review, verification, validation, changes.
- **Purchasing:** evaluation and selection of suppliers, verification of purchased product.
- **Control of nonconforming product:** identify, segregate, disposition (rework, scrap, concession).
- **Corrective action** (remove the cause of an existing nonconformity) and **preventive action** (remove cause of a potential one).
- **Calibration** of measuring equipment traceable to national standards.
- **Training, identification and traceability, handling, storage, packaging, delivery.**

ISO 9001:2015 (new) - quick note

- High level structure (Annex SL) with 10 clauses: Scope, References, Terms, **Context of the organisation,** Leadership, Planning **(risk based thinking),** Support, Operation, Performance evaluation,

Improvement.

- No mandatory quality manual; "documented information"; 7 principles (the supplier principle becomes "relationship management", the system approach is merged with process approach, "evidence based decisions").

3. Implementation of Quality System



Steps for implementing ISO 9001

1. **Top management commitment** and decision; appoint the **management representative (MR).**
2. Form the **steering committee** and project team; make an implementation plan with timeline.
3. **Awareness and training** of all employees (basics of ISO, internal auditor training).
4. **Gap analysis / initial review:** compare existing practice with the standard; identify the gaps.
5. **Document the system:** quality manual, procedures, work instructions, formats; define processes.
6. **Implement** the system - practice what is documented; maintain records; train as per the documents.
7. **Internal audits** by trained auditors; corrective actions; **management review.**
8. Choose and apply to an accredited **certification body; pre-assessment** (optional).
9. **Stage 1 (document review) and Stage 2 (on-site) audit** by the registrar.

10. Close non-conformities -> **certificate** (valid 3 years); **surveillance audits** every 6 or 12 months; recertification after 3 years.
- Time taken: 6 to 12 months; cost: consultant, training, certification fees.

Success factors
- Real commitment (not just a certificate), involvement of employees, simple documents (do not over-document), proper training, use of consultants only as guides, regular audits and review, link with business goals.

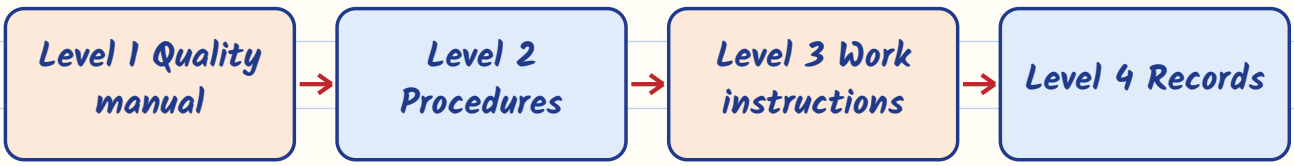
Common pitfalls
- Treating ISO as paperwork; copying other's manual; lack of ownership; poor internal audits; certificate and then relaxation.

Benefits of implementation
- Improved processes and customer satisfaction, reduced rejections, system discipline, better supplier control, marketing advantage and global acceptability.

4. Documentation

> Documentation provides **evidence** that the system is defined, understood and implemented - **"Say what you do, do what you say, prove it"** (record it).

Four levels (pyramid) of documents



- **Level 1 - Quality Manual:** describes the QMS - scope, quality policy and objectives, organisation chart, summary of procedures, process interaction. (Short, 20-40 pages; policy for "what and why".)

- **Level 2 – Procedures (SOPs): who, what, when, where** – the way activities are carried out across departments (e.g., purchase, internal audit, control of nonconforming product, corrective action, control of documents, control of records – the six mandatory procedures of ISO 9001:2000).
- **Level 3 – Work instructions, specifications, drawings, standards: how** the task is done at the work place (step-by-step; check lists).
- **Level 4 – Records, formats, forms, reports: evidence** of the activities and results – inspection reports, audit reports, calibration records, training records, minutes of management review.

Control of documents

- Approval before issue; review and update; identification of revision status; availability at points of use; legible; **obsolete documents** removed or marked; external origin documents controlled; distribution list.

Control of records

- Identification, storage, protection, retrieval, **retention time** and disposition.

Good documentation tips

- Simple, clear language; use flowcharts; reflect the actual practice; involve users in writing; minimise the paperwork; use IT (document management systems).

Document	Question it answers
Quality manual	What is our policy?
Procedure	Who does what and when?
Work instruction	How is it done?
Record	What was done / what was the result?

5. Quality Auditing

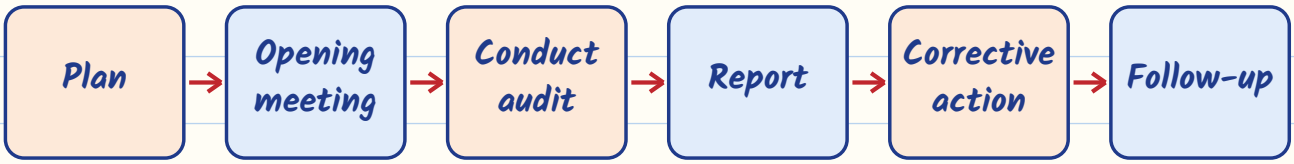
Audit: a systematic, independent and documented process for obtaining audit evidence and evaluating it objectively to determine the extent to which the **audit criteria** are fulfilled. (ISO 19011; ISO 9000:2005)

- Purpose: verify conformance with the standard and procedures, effectiveness of the system, identify improvement areas, prepare for certification, give confidence.

Types of audit

- **First party (internal) audit:** organisation audits itself, by its own trained auditors. Required by ISO 9001 at planned intervals.
- **Second party audit:** customer audits its supplier (vendor audit).
- **Third party audit:** by an independent external body (certification, surveillance audits).
- Also: system audit, process audit, **product audit**, compliance audit.

The audit process (phases)



1. **Audit planning:** scope, criteria, schedule, team (lead auditor), checklist preparation, notice to the auditee.
2. **Opening meeting** - purpose, scope, programme, and the method.
3. **Execution:** collecting evidence by interviewing, observation and examining documents and records; compare with criteria; record findings.
4. **Findings: Conformity, Non-conformity (major / minor),** observation / opportunity for improvement (OFI).
5. **Closing meeting** - present findings; agree on corrective action

timelines.

6. **Audit report** – prepared and distributed by the lead auditor.
7. **Corrective action and follow-up** by the auditee; verification of the effectiveness; closure.

- **Major NC:** absence or total breakdown of a system element/requirement; **Minor NC:** single lapse or isolated failure.

Qualities of an auditor

- Independent and objective, honest and ethical; knowledgeable of the standard and the process; good listener and communicator; observant; tactful and diplomatic; analytical; fair in reporting – "audit the system, not the person".

Audit checklist and records

- Prepared in advance with questions linked to the clause; kept as a record; helps cover all points.

Benefits of audit

- Early detection of the problems; compliance; employee awareness; basis for management review.

6. QS 9000

QS-9000 was the quality system requirement for suppliers to the automotive industry, developed by the **Chrysler, Ford and General Motors** Supplier Quality Requirements Task Force (1994). It was **ISO 9001:1994 + automotive specific requirements** (replaced later by **ISO/TS 16949** in 2002 and IATF 16949:2016).

- Aim: to **harmonise** the different supplier quality requirements of the Big Three and reduce duplication (one system for all three customers).
- Emphasis: **defect prevention,** reduction of variation and waste in the supply chain, **continuous improvement.**

Structure of QS 9000

- **Section I:** ISO 9001 requirements + additional automotive requirements (20 elements).
- **Section II: Sector specific requirements** (e.g., tooling and equipment suppliers; Ford, GM, Chrysler specific).
- **Section III: Customer specific requirements.**

Reference manuals (core tools)

- **APQP** - Advanced Product Quality Planning and Control Plan
- **FMEA** - Failure Mode and Effects Analysis
- **MSA** - Measurement System Analysis
- **SPC** - Statistical Process Control
- **PPAP** - Production Part Approval Process



Additional requirements (examples)

- Continuous improvement (quality and productivity), manufacturing capability, **process capability (Cpk >= 1.33)**, preventive maintenance, **tooling management, production part approval, design for manufacture, customer satisfaction,** supplier development, **business planning and benchmarking, bench marking, employees' empowerment.**

ISO 9001	QS 9000
General for any industry	Automotive supplier specific
System assurance	System + prevention + continuous improvement
Optional tools	Mandatory core tools (APQP, FMEA, SPC...)
Certified by registrar	Certified by an accredited registrar, recognised by the Big 3

7. ISO 14000 - Environmental Management System (EMS)

Concept

ISO 14000 is a family of standards on **environmental management** by ISO TC 207 (first issued 1996; **ISO 14001:2015** is the requirements standard). It helps an organisation **minimise the harmful effects on the environment** caused by its activities and achieve continual improvement of environmental performance.

- Based on the **PDCA:** Plan (environmental aspects, objectives), Do (implement), Check (monitor), Act (review and improve).
- Applies to all types and sizes of organisations; not a performance standard - system standard; **certifiable (14001 only).**
- Environmental aspect = element of activities that can interact with the environment (emission, effluent, waste, energy use); **Environmental impact** = the change in the environment.

The ISO 14000 family

Standard	Subject
ISO 14001	EMS requirements with guidance
ISO 14004	EMS guidelines
ISO 14010-14012 / 19011	Environmental auditing
ISO 14020	Environmental labels and declarations
ISO 14030	Environmental performance evaluation
ISO 14040	Life cycle assessment (LCA)

Requirements of ISO 14001 (core elements)

1. **Environmental policy** by top management - commitment to compliance, prevention of pollution and continual improvement.
2. **Planning:** environmental aspects and significant impacts; legal and other requirements; objectives, targets and programmes.

3. **Implementation and operation:** resources, roles and responsibilities, competence and training, communication, documentation, document control, operational control, emergency preparedness and response.
4. **Checking:** monitoring and measurement, evaluation of compliance, nonconformity and corrective action, control of records, **internal audit.**
5. **Management review** and continual improvement.



Benefits of ISO 14000

- Legal compliance and lower risk of fines; reduced consumption of energy, water and raw materials; waste reduction - cost saving.
- Better public image and community relations; marketing advantage and **green** credentials; access to global market and supply chains requiring EMS.
- Easier approval from regulators; lower insurance; employee morale; improved environmental performance; sustainability.

Implementation steps

- Commitment -> initial environmental review -> policy -> aspects and legal register -> objectives/programme -> documentation -> training -> operate -> internal audit -> management review -> certification.

ISO 9001 vs ISO 14001

Point	ISO 9001	ISO 14001
Focus	Customer / product quality	Environment / pollution prevention

Aim	Customer satisfaction	Environmental performance, compliance
Interested party	Customer	Society, regulators, community

8. Case Studies

Case 1 - ISO 9001 in a small engineering unit

- Problem: high rejection (8%), delivery delays, customer audits repeated.
- Action: gap analysis, SOPs, internal audits, training of 40 employees, supplier rating, corrective action system.
- Result: certified in 9 months; rejections down to 2%; delivery on time rose from 70% to 95%; orders increased with export customers.

Case 2 - QS 9000 at an automotive supplier (India)

- A component maker for Ford and GM implemented APQP, FMEA, SPC and PPAP; Cpk improved from 0.9 to 1.5; ppm fell from 4000 to under 100; won the Customer's supplier award.

Case 3 - ISO 14001 at a cement/paper plant

- Identified dust, energy and waste as significant aspects; installed bag filters, recycled waste heat and water; reduction in energy cost by 12%, zero legal violations; improved relations with the nearby villagers.

Case 4 - Service sector: hospital with ISO 9001

- Documented patient processes, checklists, audit system; lowered waiting time and infection rate; better patient feedback and NABH accreditation readiness.

Case 5 - Integrated management system (IMS)

- Companies combine ISO 9001 + 14001 + 45001 into one system to save documents and audits.

> How to write a case answer: Background -> Problem -> System applied -> Result -> Lesson.

Important Exam Questions (Unit V)

Short answer

- Why is ISO 9000 needed? List the 8 principles of ISO 9000:2000.
- Mention the four levels of documentation.
- Differentiate between internal and external audits.
- What are the core manuals of QS 9000?
- State any four benefits of ISO 14000.

Long answer

1. Explain the elements of ISO 9001:2000 quality system.
2. Explain the steps for implementing a quality system. How is certification obtained?
3. Explain the documentation of the quality system with a pyramid.
4. Describe the quality audit process and the types of audit.
5. Explain QS 9000 and its requirements.
6. Explain the ISO 14000 concept, requirements and benefits.
7. Discuss a case study on implementation of ISO 9000 / ISO 14000.